

OLBIA COSTA SMERALDA, Italy

WMO: 165310

Lat: 40.899N

Lon: 9.518E

Elev: 37

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	33.9	36.3	25.2	19.3	44.8	28.4	22.4	43.8	30.8	55.5	27.6	55.0	4.0	220	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	93.8	72.3	91.4	71.8	89.3	71.4	77.4	86.7	75.8	85.6	74.4	84.3	11.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	131.6	82.0	73.1	123.0	80.9	71.3	115.7	80.3	40.8	86.6	39.2	85.6	37.8	84.6	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	22.9	20.1	DB	29.2	99.9	2.0	3.0	27.7	102.1	26.5	103.8	25.4	105.6	23.9	107.8	
			WB	28.2	79.7	2.3	3.1	26.5	81.9	25.2	83.7	23.9	85.4	22.2	87.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	49.6	49.6	53.2	57.5	64.5	72.6	78.2	78.2	71.4	65.1	57.0	51.6	(6)
	DBStd	11.26	4.67	4.75	4.70	4.15	4.45	4.94	3.86	3.68	4.51	4.53	4.90	4.90	(7)
	HDD50	188	63	59	20	2	0	0	0	0	0	0	6	38	(8)
	HDD65	2285	478	431	365	226	64	4	0	0	4	54	244	415	(9)
	CDD50	4734	51	48	120	227	449	678	875	873	643	467	215	88	(10)
	CDD65	1354	0	0	0	1	47	232	410	408	198	56	2	0	(11)
	CDH74	13116	0	0	3	17	401	2265	4471	4282	1398	272	7	0	(12)
	CDH80	5184	0	0	0	0	68	806	2037	1897	346	30	0	0	(13)
(14) Wind	WSAvg	8.4	8.7	8.8	9.8	8.8	8.6	8.4	9.1	8.1	7.9	7.0	7.5	8.5	(14)
(15) Precipitation	PrecAvg	21.1	2.0	2.7	2.4	2.0	1.2	0.9	0.2	0.7	1.1	2.6	2.4	3.9	(15)
	PrecMax	28.1	5.3	9.2	4.8	6.7	5.3	2.6	1.1	2.0	2.7	7.8	6.0	14.8	(16)
	PrecMin	15.2	0.1	0.1	0.1	0.4	0.0	0.0	0.0	0.0	0.1	0.2	0.2	0.2	(17)
	PrecStd	3.9	1.4	2.0	1.4	1.5	1.1	0.9	0.3	0.7	0.9	2.4	1.6	3.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	66.3	71.9	76.6	86.2	95.1	98.0	98.5	89.9	83.9	74.9	67.6	(19)
		MCWB	56.6	55.9	58.1	60.8	65.5	70.3	72.6	73.1	70.9	69.8	62.8	58.6	(20)
	2%	DB	62.4	62.9	68.0	72.7	81.0	91.1	93.6	94.3	86.2	79.2	71.2	64.5	(21)
		MCWB	55.2	54.1	55.7	59.7	63.9	70.5	71.7	73.9	70.4	67.7	62.3	57.4	(22)
	5%	DB	60.4	60.8	64.8	69.8	78.5	87.6	91.4	91.4	84.1	77.0	68.2	62.5	(23)
		MCWB	54.1	53.1	54.8	58.5	63.4	69.1	71.3	72.9	69.8	66.7	60.9	56.4	(24)
	10%	DB	58.6	58.8	62.7	67.7	75.4	84.4	89.4	88.8	81.6	74.8	66.0	60.6	(25)
		MCWB	52.6	51.8	54.5	57.7	62.6	68.0	71.3	72.3	69.0	65.6	59.5	55.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	57.6	60.6	63.8	68.8	76.2	79.1	80.7	76.5	72.7	65.7	60.5	(27)
		MCDB	62.9	63.6	68.3	71.5	78.0	87.2	88.6	87.6	82.7	79.0	70.2	64.8	(28)
	2%	WB	56.6	55.5	58.3	61.8	67.0	73.5	76.9	78.2	74.1	70.1	63.7	58.8	(29)
		MCDB	60.9	61.2	64.4	69.0	76.1	84.7	87.1	86.6	81.6	75.9	68.7	63.0	(30)
	5%	WB	54.7	54.1	56.7	60.3	65.5	71.8	75.4	76.4	72.4	68.5	62.0	57.2	(31)
		MCDB	59.3	59.4	62.5	67.1	74.4	82.4	85.7	86.0	80.1	74.5	66.8	61.3	(32)
	10%	WB	52.9	52.6	55.3	58.9	64.1	70.1	73.9	74.9	70.8	66.9	60.1	55.6	(33)
		MCDB	57.5	57.6	61.1	65.1	72.6	80.7	84.3	85.1	78.5	73.0	65.0	59.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	16.3	17.4	18.3	19.8	21.1	21.2	20.9	18.7	18.0	15.5	14.8	(35)
		MCDBR	17.1	19.2	21.0	22.5	24.0	25.3	24.3	25.2	21.1	20.8	18.8	16.5	(36)
	5% WB	MCWBR	11.8	12.4	12.6	12.6	11.5	11.4	10.7	12.7	10.4	11.7	11.9	11.2	(37)
		MCWBR	16.2	17.9	18.9	20.1	21.5	22.9	22.6	23.1	20.5	19.9	17.7	15.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.336	0.354	0.384	0.417	0.430	0.412	0.417	0.414	0.410	0.392	0.368	0.340	(40)
		taud	2.473	2.409	2.335	2.255	2.259	2.349	2.346	2.364	2.371	2.398	2.407	2.456	(41)
	Ebn at Noon		259	271	274	272	270	274	272	269	262	253	243	246	(42)
		Edh at Noon	25	30	36	42	43	39	39	37	35	30	26	23	(43)
(44) All-Sky Solar Radiation	RadAvg		576	851	1247	1629	1974	2275	2334	2025	1504	1049	652	517	(44)
	RadStd		61	84	136	117	167	99	94	109	62	91	52	53	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +0.56	(e) N/A	(f) N/A	(g) N/A	(h) -1.22	(i) -1.69	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page